

Last name _____

First name _____

LARSON—MATH 310—HOMEWORK WORKSHEET 04

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Problems

1. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is the block vector $\vec{w} = \begin{bmatrix} \vec{v} \\ \vec{u} \end{bmatrix}$
2. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} + \vec{u}$?
3. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} - \vec{u}$?
4. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$, what is $4\vec{v}$?
5. What is a *linear combination*? (Explain and give an example).
6. Find the *inner product* of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.
7. (**Notation**) What is $\mathbb{R}^3$?
8. Consider the function $f(x, y) = 2x + 3y$. Check that f is *additive*; that is, show:
$$\text{For any pairs } (x_1, y_1) \text{ and } (x_2, y_2), f((x_1, y_1) + (x_2, y_2)) = f(x_1, y_1) + f(x_2, y_2).$$
9. Consider the function $f(x, y) = 2x + 3y$. Check that f is *homogenous*; that is, show:
$$\text{For any pair } (x, y) \text{ and scalar } \alpha, f(\alpha(x, y)) = \alpha f(x, y).$$
10. Explain why the function $f(x, y) = 2x + 3y$ is linear.
11. How can we write the linear function $f(x, y) = 2x + 3y$ as an inner product?

(over)

12. Explain why all linear functions go through the origin? (If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear function, show $f(\vec{0}) = 0$).
13. What is an *affine function*? (Explain and give an example).
14. Suppose $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function? Maybe its an affine function? Here's all we know:

$$\phi(1, 1, 0) = -1$$

$$\phi(-1, 1, 1) = 1$$

$$\phi(1, -1, 1) = 1$$

15. **Transforming between two encodings for Boolean vectors.** A Boolean 4-vector is one for which all entries are either 0 or 1. Such vectors are used to encode whether each of n conditions holds, with $a_i = 1$ meaning that condition i holds. Another common encoding of the same information uses the two values -1 and $+1$ for the entries.

For example the Boolean vector $(0, 1, 1, 0)$ would be written using this alternative encoding as $(-1, +1, +1, -1)$.

Suppose that x is a Boolean vector with entries that are 0 or 1, and y is a vector encoding the same information using the values -1 and $+1$. Express y in terms of x .

Compute examples. **Show all your reasoning.**